

Class 08: Breast Cancer Mini Project

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Background

The goal of this mini-project is for you to explore a complete analysis using the unsupervised learning techniques covered in class

Today we will analyze a biopsy data-set from fine needle aspiration (FNA) of a breast mass.

Data Import

The data is made available as a CSV file for download.

```
wisc.df <- read.csv("WisconsinCancer.csv", row.names=1)
```

```
head(wisc.df, 3)
```

	diagnosis	radius_mean	texture_mean	perimeter_mean	area_mean	
842302	M	17.99	10.38	122.8	1001	
842517	M	20.57	17.77	132.9	1326	
84300903	M	19.69	21.25	130.0	1203	
		smoothness_mean	compactness_mean	concavity_mean	concave.points_mean	
842302		0.11840	0.27760	0.3001	0.14710	
842517		0.08474	0.07864	0.0869	0.07017	
84300903		0.10960	0.15990	0.1974	0.12790	
		symmetry_mean	fractal_dimension_mean	radius_se	texture_se	perimeter_se
842302		0.2419	0.07871	1.0950	0.9053	8.589
842517		0.1812	0.05667	0.5435	0.7339	3.398
84300903		0.2069	0.05999	0.7456	0.7869	4.585
		area_se	smoothness_se	compactness_se	concavity_se	concave.points_se
842302		153.40	0.006399	0.04904	0.05373	0.01587
842517		74.08	0.005225	0.01308	0.01860	0.01340
84300903		94.03	0.006150	0.04006	0.03832	0.02058
		symmetry_se	fractal_dimension_se	radius_worst	texture_worst	
842302		0.03003	0.006193	25.38	17.33	
842517		0.01389	0.003532	24.99	23.41	
84300903		0.02250	0.004571	23.57	25.53	
		perimeter_worst	area_worst	smoothness_worst	compactness_worst	
842302		184.6	2019	0.1622	0.6656	
842517		158.8	1956	0.1238	0.1866	
84300903		152.5	1709	0.1444	0.4245	
		concavity_worst	concave.points_worst	symmetry_worst		
842302		0.7119	0.2654	0.4601		
842517		0.2416	0.1860	0.2750		
84300903		0.4504	0.2430	0.3613		
		fractal_dimension_worst				
842302		0.11890				
842517		0.08902				
84300903		0.08758				

Make sure we remove or exclude the `diagnosis` column from the data-set that we use for further analysis - this is the expert diagnosis as either M or B:

```
wisc.data <- wisc.df[,-1]
diagnosis <- as.factor(wisc.df$diagnosis)
```

Exploratory Data Analysis

Q1: How many observations are in this dataset?

```
nrow(wisc.data)
```

```
[1] 569
```

569 patients (observations)

Q2: How many of the observations have a malignant diagnosis?

```
table(wisc.df$diagnosis)
```

```
  B    M  
357 212
```

212 malignant

Q3: How many variables/features in the data are suffixed with `_mean`?

We can use the `grep()` function to help us

```
length(grep("_mean", colnames(wisc.data)))
```

```
[1] 10
```

Principal Component Analysis (PCA)

We need to scale our data before PCA with the `scale=TRUE` argument to `prcomp()`.

```
wisc.pr <- prcomp(wisc.data, scale = TRUE)  
summary(wisc.pr)
```

Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	3.6444	2.3857	1.67867	1.40735	1.28403	1.09880	0.82172
Proportion of Variance	0.4427	0.1897	0.09393	0.06602	0.05496	0.04025	0.02251
Cumulative Proportion	0.4427	0.6324	0.72636	0.79239	0.84734	0.88759	0.91010

	PC8	PC9	PC10	PC11	PC12	PC13	PC14
Standard deviation	0.69037	0.6457	0.59219	0.5421	0.51104	0.49128	0.39624
Proportion of Variance	0.01589	0.0139	0.01169	0.0098	0.00871	0.00805	0.00523

Cumulative Proportion	0.92598	0.9399	0.95157	0.9614	0.97007	0.97812	0.98335
	PC15	PC16	PC17	PC18	PC19	PC20	PC21
Standard deviation	0.30681	0.28260	0.24372	0.22939	0.22244	0.17652	0.1731
Proportion of Variance	0.00314	0.00266	0.00198	0.00175	0.00165	0.00104	0.0010
Cumulative Proportion	0.98649	0.98915	0.99113	0.99288	0.99453	0.99557	0.9966
	PC22	PC23	PC24	PC25	PC26	PC27	PC28
Standard deviation	0.16565	0.15602	0.1344	0.12442	0.09043	0.08307	0.03987
Proportion of Variance	0.00091	0.00081	0.0006	0.00052	0.00027	0.00023	0.00005
Cumulative Proportion	0.99749	0.99830	0.9989	0.99942	0.99969	0.99992	0.99997
	PC29	PC30					
Standard deviation	0.02736	0.01153					
Proportion of Variance	0.00002	0.00000					
Cumulative Proportion	1.00000	1.00000					

Q4: From your results, what proportion of the original variance is captured by the first principal component (PC1)?

0.4427 (44.27%)

Q5: How many principal components (PCs) are required to describe at least 70% of the original variance in the data?

At least 3 PCs

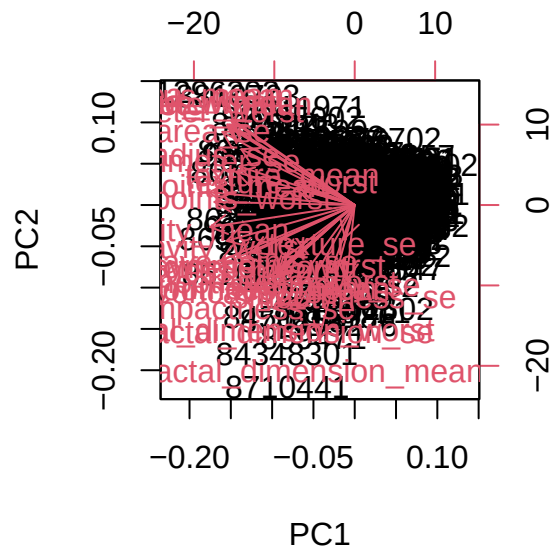
Q6: How many principal components (PCs) are required to describe at least 90% of the original variance in the data?

At least 7 PCs

Interpreting PCA Results

Now you will use some visualizations to better understand your PCA model. A common visualization for PCA results is the so-called biplot.

```
biplot(wisc.pr)
```



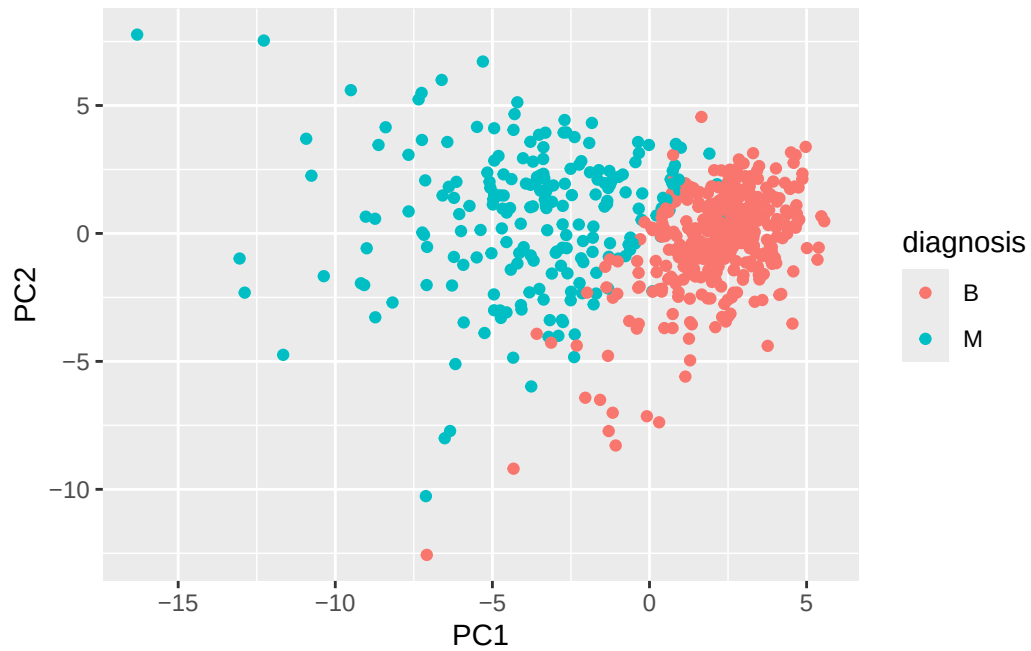
Q7: What stands out to you about this plot? Is it easy or difficult to understand? Why?

This plot is very difficult to understand because there are way too many data points/values to visualize on the graph, everything just blends together. The column names are red, but the rest of the numerical values all blend together.

Lets see the “PC score plot”

```
library(ggplot2)

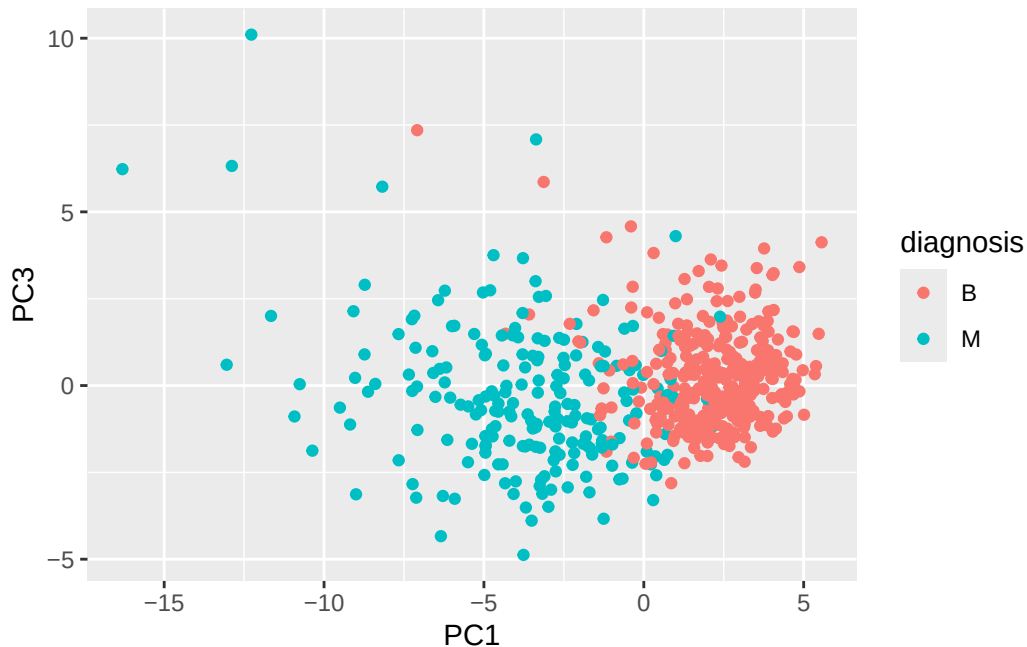
ggplot(wisc.pr$x) +
  aes(PC1, PC2, col = diagnosis) +
  geom_point()
```



Q8: Generate a similar plot for principal components 1 and 3. What do you notice about these plots?

Both these plots show that PC1 captures a separation of malignant and benign samples.

```
ggplot(wisc.pr$x) +  
  aes(PC1, PC3, col=diagnosis) +  
  geom_point()
```



Variance Explained

A scree plot shows how much variance each PC captures. We typically look for an “elbow” — a point where adding more PCs gives diminishing returns.

Calculate the variance of each principal component by squaring the `sdev` component of `wisc.pr` (i.e. `wisc.pr$sdev^2`). Save the result as an object called `pr.var`.

```
# Calculate variance of each component
pr.var <- wisc.pr$sdev^2
head(pr.var)
```

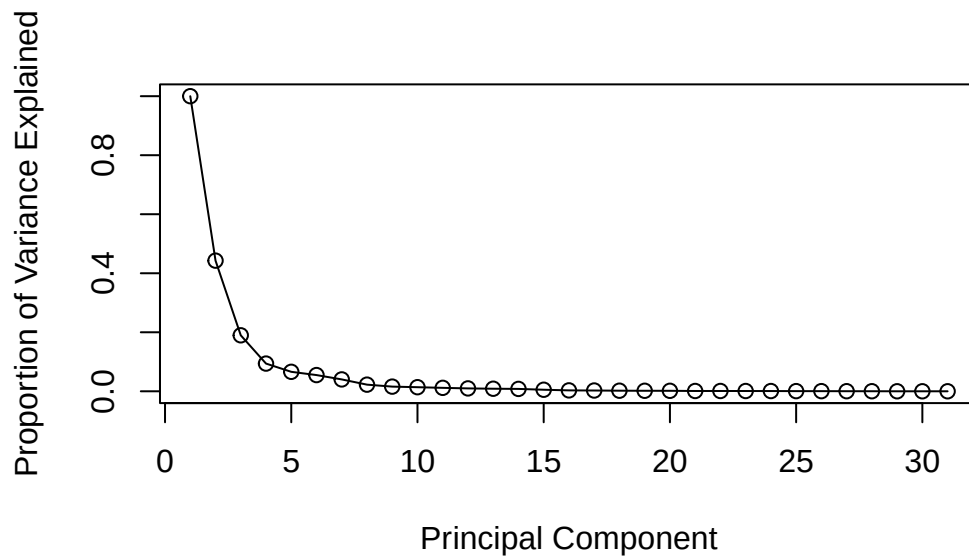
```
[1] 13.281608  5.691355  2.817949  1.980640  1.648731  1.207357
```

Calculate the variance explained by each principal component by dividing by the total variance explained of all principal components.

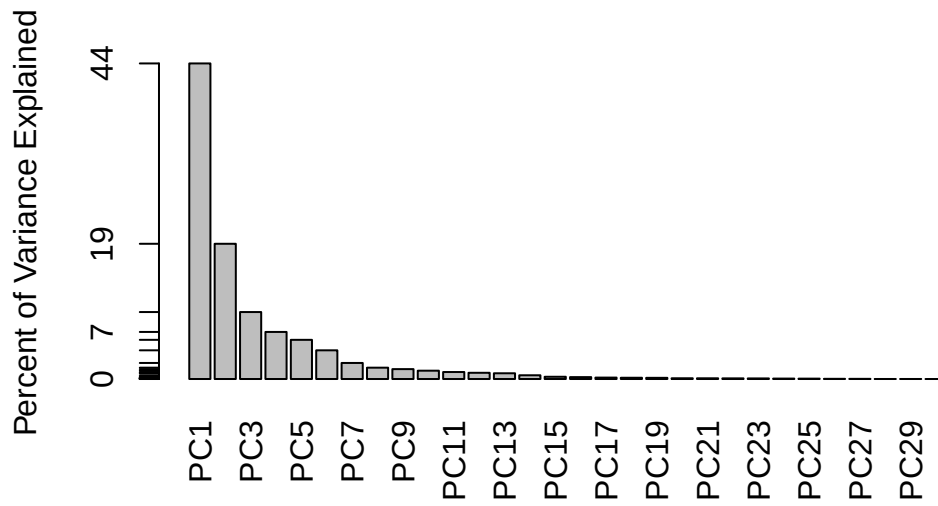
```
# Variance explained by each principal component: pve
pve <- pr.var / sum(pr.var)

# Plot variance explained for each principal component
```

```
plot(c(1,pve), xlab = "Principal Component",
     ylab = "Proportion of Variance Explained",
     ylim = c(0, 1), type = "o")
```



```
# Alternative scree plot of the same data, note data driven y-axis
barplot(pve, ylab = "Percent of Variance Explained",
        names.arg=paste0("PC",1:length(pve)), las=2, axes = FALSE)
axis(2, at=pve, labels=round(pve,2)*100 )
```

Communicating PCA Results

The loading vector (`wisc.pr$rotation`) tells us which original measurements contribute most to each PC.

Q9: For the first principal component, what is the component of the loading vector (i.e. `wisc.pr$rotation[,1]`) for the feature `concave.points_mean`? This tells us how much this original feature contributes to the first PC. Are there any features with larger contributions than this one?

```
wisc.pr$rotation["concave.points_mean", 1]
```

```
[1] -0.2608538
```

```
sort(abs(wisc.pr$rotation[,1]))
```

smoothness_se	texture_se	symmetry_se
0.01453145	0.01742803	0.04249842
fractal_dimension_mean	fractal_dimension_se	texture_mean
0.06436335	0.10256832	0.10372458
texture_worst	symmetry_worst	smoothness_worst

0.10446933	0.12290456	0.12795256
fractal_dimension_worst	symmetry_mean	smoothness_mean
0.13178394	0.13816696	0.14258969
concavity_se	compactness_se	concave.points_se
0.15358979	0.17039345	0.18341740
area_se	radius_se	compactness_worst
0.20286964	0.20597878	0.21009588
perimeter_se	radius_mean	area_mean
0.21132592	0.21890244	0.22099499
area_worst	perimeter_mean	radius_worst
0.22487053	0.22753729	0.22799663
concavity_worst	perimeter_worst	compactness_mean
0.22876753	0.23663968	0.23928535
concave.points_worst	concavity_mean	concave.points_mean
0.25088597	0.25840048	0.26085376

The component of the loading vector is -2.608538, there aren't any features with larger contributions

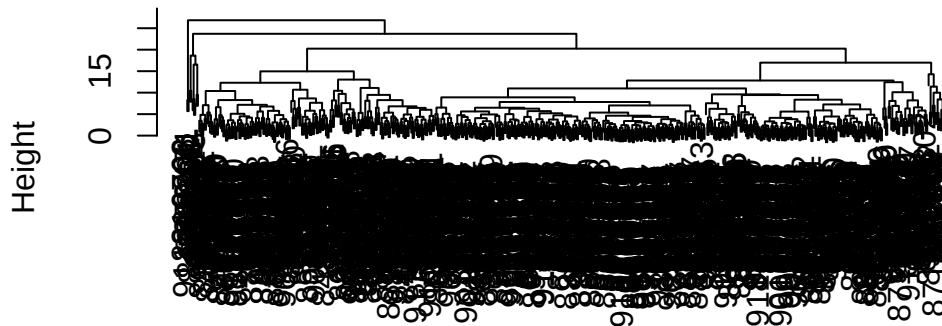
Hierarchical clustering

As part of the preparation for hierarchical clustering, the distance between all pairs of observations needs to be calculated. This “distance matrix” will be the input for the `hclust()` function.

First scale the `wisc.data` data and assign the result to `data.scaled`.

```
wisc.hclust <- hclust(dist(scale(wisc.data)))
plot(wisc.hclust)
```

Cluster Dendrogram

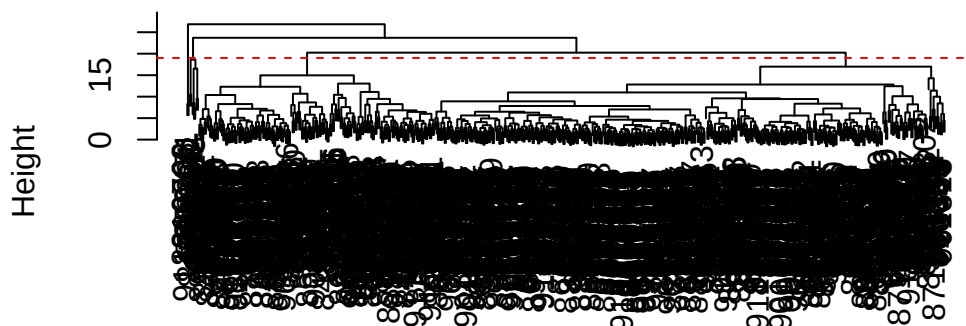


```
dist(scale(wisc.data))  
hclust (*, "complete")
```

Q10: Using the `plot()` and `abline()` functions, what is the height at which the clustering model has 4 clusters?

```
plot(wisc.hclust)  
abline(h = 19, col = "red", lty = 2)
```

Cluster Dendrogram



```
dist(scale(wisc.data))
hclust(*, "complete")
```

The clustering model has 4 clusters at $h = 19$.

Use `cutree()` to cut the tree so that it has 4 clusters. Assign the output to the variable `wisc.hclust.clusters`.

```
wisc.hclust.clusters <- cutree(wisc.hclust, h = 19)
```

```
table(wisc.hclust.clusters, diagnosis)
```

	diagnosis	
wisc.hclust.clusters	B	M
1	12	165
2	2	5
3	343	40
4	0	2

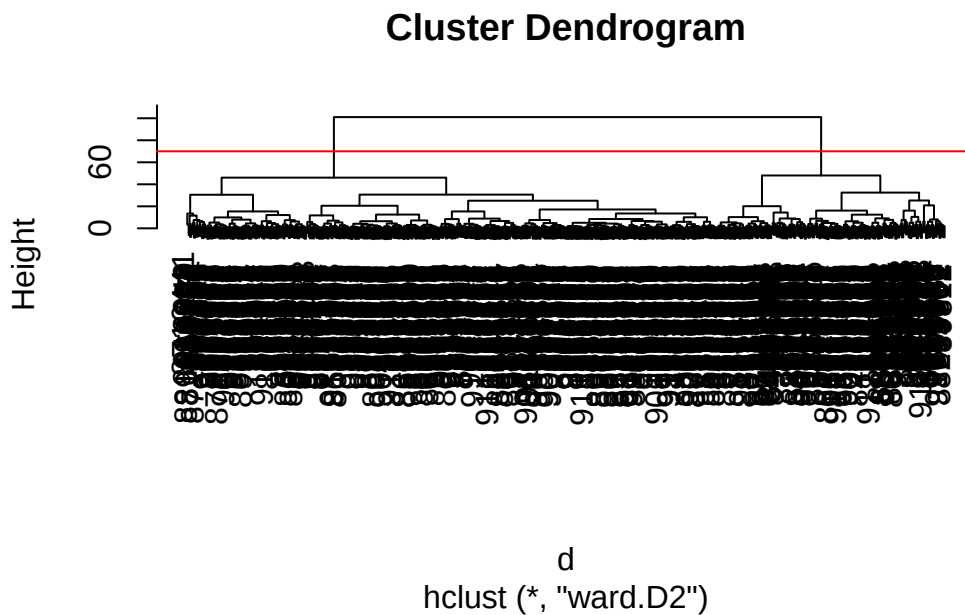
As we discussed in our last class videos there are number of different “methods” we can use to combine points during the hierarchical clustering procedure. These include “single”, “complete”, “average” and (my favorite) “ward.D2”

Q12: Which method gives your favorite results for the same data.dist dataset?
Explain your reasoning.

I prefer **ward.D2** because it produces more compact and well-separated clusters by minimizing within-cluster variance. This makes the dendrogram easier to interpret compared to methods like single linkage, which can create chain-like clusters.

Combining Methods

```
d <- dist(wisc.pr$x[,1:4])
wisc.pr.hclust <- hclust(d, method = "ward.D2")
plot(wisc.pr.hclust)
abline(h=70, col = "red")
```



```
grps <- cutree(wisc.pr.hclust, h=70)
table(grps)
```

```
grps
  1   2
171 398
```

How does this clustering `grps` correspond to the expert diagnosis?

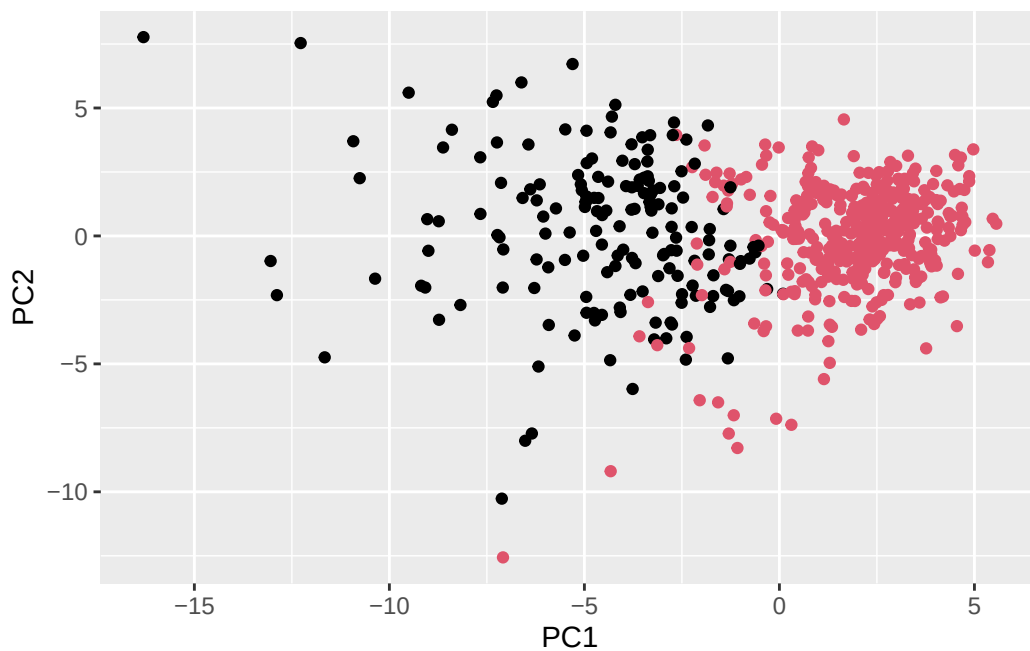
```
table(diagnosis)
```

```
diagnosis
  B    M
357 212
```

```
table(diagnosis, grps)
```

```
      grps
diagnosis  1   2
      B    6 351
      M   165 47
```

```
ggplot(wisc.pr$x) +
  aes(PC1, PC2) +
  geom_point(col=grps)
```



```
wisc.pr.hclust.clusters <- cutree(wisc.pr.hclust, k = 2)
```

Using `table()`, compare the results from your new hierarchical clustering model with the actual diagnoses.

Q13: How well does the newly created hclust model with two clusters separate out the two “M” and “B” diagnoses?

```
table(wisc.pr.hclust.clusters, diagnosis)
```

	diagnosis	
wisc.pr.hclust.clusters	B	M
1	6	165
2	351	47

The 2-cluster model separates the data fairly well: one cluster is mostly malignant (165 M vs 6 B), and the other is mostly benign (351 B vs 47 M), though some misclassification remains.

Q14: How well do the hierarchical clustering models you created in the previous sections (i.e. without first doing PCA) do in terms of separating the diagnoses? Again, use the table() function to compare the output of each model (wisc.hclust.clusters and wisc.pr.hclust.clusters) with the vector containing the actual diagnoses.

```
table(wisc.hclust.clusters, diagnosis)
```

	diagnosis	
wisc.hclust.clusters	B	M
1	12	165
2	2	5
3	343	40
4	0	2

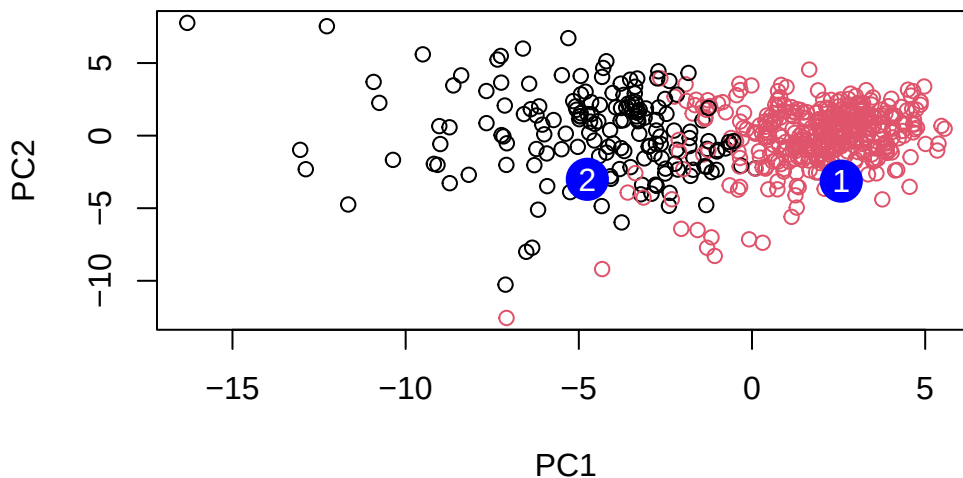
Without PCA, the clustering is less clean because the diagnoses are spread across multiple clusters instead of being clearly separated into two groups. Even if one cluster has fewer errors (e.g., 40 M), the overall separation is worse than the PCA-based 2-cluster model

##Prediction

```
#url <- "new_samples.csv"
url <- "https://tinyurl.com/new-samples-CSV"
new <- read.csv(url)
npc <- predict(wisc.pr, newdata=new)
npc
```

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
[1,]	2.576616	-3.135913	1.3990492	-0.7631950	2.781648	-0.8150185	-0.3959098
[2,]	-4.754928	-3.009033	-0.1660946	-0.6052952	-1.140698	-1.2189945	0.8193031
	PC8	PC9	PC10	PC11	PC12	PC13	PC14
[1,]	-0.2307350	0.1029569	-0.9272861	0.3411457	0.375921	0.1610764	1.187882
[2,]	-0.3307423	0.5281896	-0.4855301	0.7173233	-1.185917	0.5893856	0.303029
	PC15	PC16	PC17	PC18	PC19	PC20	
[1,]	0.3216974	-0.1743616	-0.07875393	-0.11207028	-0.08802955	-0.2495216	
[2,]	0.1299153	0.1448061	-0.40509706	0.06565549	0.25591230	-0.4289500	
	PC21	PC22	PC23	PC24	PC25	PC26	
[1,]	0.1228233	0.09358453	0.08347651	0.1223396	0.02124121	0.078884581	
[2,]	-0.1224776	0.01732146	0.06316631	-0.2338618	-0.20755948	-0.009833238	
	PC27	PC28	PC29	PC30			
[1,]	0.220199544	-0.02946023	-0.015620933	0.005269029			
[2,]	-0.001134152	0.09638361	0.002795349	-0.019015820			

```
plot(wisc.pr$x[,1:2], col=grps)
points(npc[,1], npc[,2], col="blue", pch=16, cex=3)
text(npc[,1], npc[,2], c(1,2), col="white")
```



Q16: Which of these new patients should we prioritize for follow up based on your results?

Patient 2, they are in the “malignant” group.